MIL-M-17522D(SHIPS)
16 October 1967
SUPERSEDING
MIL-M-17522C(SHIPS)
21 March 1967
(See 6.5)

MILITARY SPECIFICATION

MASK, PROTECTIVE, ND MARK V

1. SCOPE

1.1 This specification covers a protective mask for respiratory protection of personnel aboard ships against radiological bacteriological and chemical warfare.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
- PPP-B-566 - Boxes, Folding.
- PPP-B-636 - Box, Fiberboard.
- PPP-B-665 - Boxes, Paperboard, Metal, Stayed (Including Stay Material).
- PPP-B-676 - Boxes, Setup.

MILITARY
- MIL-P-116 - Preservation, Methods of.
- MIL-R-880 - Rubber, Medium-Soft (for Airport, Air-tight and Watertight - Door; and Hatch Gaskets).
- MIL-R-900 - Rubber Gasket Material, 45 Durometer Hardness, for Moderately Low-Temperature Service.
- MIL-D-10132 - Disc, Outlet Valve.
- MIL-A-15042 - Antifogging Compound, Eye Protective Equipment.
- MIL-M-17921 - Mask, Protective, ND Mark V Technical Requirements for.
- MIL-T-19102 - Test Procedures for Mask, Gas ND Mark V.
- MIL-C-24314 - Canister, ND-C-2.

STANDARDS

FEDERAL
- FED-STD-601 - Rubber: Sampling and Testing.

MILITARY
- MIL-STD-8 - Dimensioning and Tolerancing.
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
- MIL-STD-758 - Packaging Procedures for Submarine Repair Parts Utilizing Transparent, Flexible, Heat Sealable Film.

DRAWINGS

MILITARY
- D-1979 - Light Transmission Device for Gas Mask Lens.
- D-1982 - Pull Device for ND Mk V Canister Holder.
- 805-1327140 - Gas Mask ND Mark V Assembly and Faceblank Sections.
- 805-1327141 - Gas Mask ND Mark V Assembly and Faceblank Sections.
- 805-1327142 - Gas Mask ND Mark V Facepiece Parts and Head Harness Details.
- 805-1327143 - Gas Mask ND Mark V Facepiece Parts and Head Harness Details.
- 805-1327144 - Gas Mask ND Mark V Speech Diaphragm Outlet Valve, Assembly and Details.
- 805-1327145 - Gas Mask ND Mark V Speech Diaphragm Outlet Valve, Assembly and Details.
- 805-1327146 - Protective Mask ND Mark V Canister Assembly and Details.
- 805-1327147 - Gas Mask ND Mark V Carrier Assembly and Details.
- 805-1327148 - Gas Mask ND Mark V Carrier Assembly and Details.
- 805-1327149 - Gas Mask ND Mark V Faceblank Core, Form and Sections.

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MILITARY (cont'd)

805-1327150 - Gas Mask ND Mark V Faceblank Core, Form and Sections.
805-1327151 - Gas Mask ND Mark V Faceblank Core, Form and Sections.
805-1327152 - Gas Mask ND Mark V Facebland Core, Form and Sections.
805-1327153 - Gas Mask ND Mark V Faceblank Cavity Molds.
805-1327154 - Gas Mask ND Mark V Faceblank Cavity Molds.
805-1327155 - Gas Mask ND Mark V Faceblank Cavity Molds.
805-1327156 - Gas Mask ND Mark V Faceblank Mold.
805-1327157 - Gas Mask ND Mark V Faceblank Mold.
805-1327158 - Gas Mask ND Mark V Faceblank Mold.
805-1327159 - Gas Mask ND Mark V Faceblank Mold.

PUBLICATIONS

MILITARY
Chemical Warfare Service Pamphlet No. 2: Part 2, Section L.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

GOVERNMENTAL
NATIONAL BUREAU OF STANDARDS HANDBOOK
H-28 - Screw-Thread-Standards for Federal Services.

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)
D 1149 - Method of Test for Accelerated Ozone Cracking of Vulcanized Rubber.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample. Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.4).

3.2 Component parts. The protective mask shall consist of a facepiece, a head harness, two canisters, a carrier, and an anti-fog kit.

3.2.1 Facepiece. The facepiece shall be made of rubber, and shall have attachments consisting of five buckles, two canister holders, two inlet valves, speech diaphragm outlet valve assembly, lens, and two air deflector tubes.

3.2.2 Canister. The canister shall conform to MIL-C-24314.

3.2.3 Head harness. The head harness shall be made of rubber.

3.2.4 Carrier. The carrier shall be made of waterproofed-mildew resistant fabric and shall be equipped with snap(s) for quick opening, straps with snaps for carrying from the shoulder or waist, reinforcing shield and lining for protecting the protective mask lens, loop for holding anti-fog kit and space for a kit of protective ointment.

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3.2.5 Anti-fog kit. The anti-fog kit shall conform to MIL-A-15042.

3.3 Material.

3.3.1 Rubber compound. The rubber compounds shall be nonsulfurblooming and contain not less than 85 percent by volume of good or standard grade plantation rubber, except as specified in table I, free from lead or mineral sulphate and not more than 3.5 percent sulphur by weight. The rubber of the finished components shall be nontoxic, when tested in accordance with 4.6.4 and shall have the physical characteristics specified in table I. It shall not deleteriously affect the color or light transmission of the lens when aged in accordance with 4.6.5.3. Toxicity tests are required only for rubber compounds used in the manufacture of the facepiece, head harness and deflector tubes. All rubber surfaces shall be free of pits, bumps, non-fills, porous areas, spongy areas, blisters, air-traps and flow cracks. Unless otherwise specified (see 6.2), small defects which do not extend in depth more than one-quarter of the thickness of the rubber may be repaired.

Table I - Physical characteristics for rubber compound.

Description	Before aging						After aging 1/	
	Tensile strength (min.)	Elongation (min.)	Permanent set (max.)	Hardness	Average tensile stress at 200 percent elongation 5/	Tensile stress at 200 percent elongation 5/	Tensile strength	Elongation
	P.s.i	Percent	Percent	Shore	P.s.i	P.s.i	Percent	Percent
Face-blank 2/ Pc1-1	3,000	675	20	35±5	150-300	150-300	-25	+20 -15
Outlet 3/ valve disc Pc3-8	3,000	675	20	35±5	150-300	150-300	-25	+20 -15
Flapper valve of. Outlet valve cover Pc3-6 4/	3,000	675	20	35±5	150-300	150-300	-25	+20
Inlet valve disc Pc2-8 4/	---	---	--	45±5	---	---	---	--
Head harness Pc2-9 6/	2,300 3,000	250 500	27 20	70±10 45±5	1,200-1,400 200-400	1,200-1,400 200-400	-26 -25	-32 +20 -15
Deflector tube Pc2-2 7/	3,000	500	20	45±5	200-400	200-400	-25	+20 -15
Diaphragm gasket Pc3-3 8/	----	---	--	45±5	----	----	---	---

1/ Characteristics after aging shall be based on minimum specification requirements.

2/ Compression set shall not exceed 55 percent when determined as specified in MIL-R-880.

3/ May be manufactured in accordance with type I of MIL-D-10132 excepting leakage and resistance requirements shall be in accordance with 3.4.1.7.2, color optional.

4/ Physical requirements not specified in table I shall conform to MIL-R-900 except for low temperatures and seal aging.

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5/ No individual test specimen shall be less than 90 percent of the minimum specified.

6/ P.s.i. 2,300 for high modulus rubber shall contain not less than 65 percent by volume of good or standard grade plantation rubber. P.s.i. 3,000 for low modulus rubber.

7/ Compression set shall conform to MIL-R-880.

8/ Physical requirements not specified in table I shall conform to MIL-R-900 except for low temperatures. For seal aging test use method 3211 of FED-STD-601.

3.3.1.1 Sun checking. The rubber compound shall contain from 0.5 to 1.0 percent sun checking compound.

3.3.1.2 Antioxidants. The rubber compound shall contain an antioxidant compound equivalent to those known commercially as American Cyanamid Corporation antioxidant 2246 or Agerite Stalite and as compounded shall be free from any materials which when in contact with the face will irritate the skin or adversely affect the light transmission of the lens.

3.3.1.3 Copper inhibitor. The rubber compound shall contain a copper inhibitor of such a material that it will not adversely affect the physical properties of the rubber.

3.3.1.4 Coloring materials. Unless otherwise specified (see 6.2), the rubber compound shall be colored a dull black by mineral pigments and carbon blacks.

3.3.2 Metals. Metals shall be as specified in the applicable drawings.

3.3.2.1 Machine threads. Machine threads shall be in accordance with H28. Dimensions and tolerances shall conform to MIL-STD-8.

3.3.3 Fabrics. All fabric, webbing and thread of the carrier shall be made water repellent and mildew resistant in accordance with 805-1327147 and 805-1327148. The color shall be haze grey (see 6.3). The cloth and webbing of any carrier shall each be taken from the same lot in order to provide a uniform color in each carrier. All fabrics shall conform to requirements of the applicable drawings.

3.3.3.1 Cloth reinforcement. The cloth for rubber reinforcement shall be woven light weight frictioned fabric.

3.3.3.2 Lens material. The lens material shall be in accordance with 805-1327142.

3.4 Design and manufacture.

3.4.1 Facepiece. The facepiece shall consist of the rubber faceblank, shown on 805-1327140 and 805-1327141 and pieces 2-1 through 2-8 and 2-10 through 2-12 shown on 805-1327143.

3.4.1.1 Faceblank. The faceblank shall be one piece, molded of rubber conforming to 3.3.1. The faceblank shall be made from a mold conforming to 805-1327149 through 805-1327159 and shall conform to 805-1327140. The mold cavity and core shall have a tolerance of plus or minus 0.010 of the mean dimension unless otherwise specified on the referenced drawings. The thickness of the rubber on the facial contact surface of the pneumatic tube shall be 0.040±.010 inch for 90 percent of the surface; the remaining 10 percent may be 0.040± .020 inch. The facial contact surface is that portion of the pneumatic tube which contacts the face of the wearer and constitutes the entire periphery of the tube exclusive of 1/8 inch width strip of each side which parallels and forms the junction to the facepiece proper. The finish on the contact surface of the pneumatic tube shall not exceed the equivalent of a 16 (grind) microinch finish over 90 percent of its total area. The finish on the remaining 10 percent of the contact surface shall not exceed the equivalent of a 32 (grind) microinch finish or provide an area across the center 3/4 inch of the pnuematic tube having a finish exceeding the equivalent of a 16 (grind) microinch finish. No repairs are allowed. The completed pneumatic tube shall be opened to the atmosphere for a sufficient time to insure pressure equilibrium and then resealed. Care shall be taken that no uncured cement or solvent remains inside of the completed pneumatic tube. The temperature of the mask and the atmospheric pressure at the time of pressure equilibration shall be within the following limits:

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Pressure (mm. Hg.)	Temperature (°C.) Minimum	Maximum
720	6.4	21.1
730	11.1	25.6
740	15.8	30.0
750	19.4	33.3
760	22.9	36.8
770	26.5	40.2
780	30.1	43.7

3.4.1.2 Marking. The marking shall be in accordance with that indicated on applicable drawings and appear in readily readable symbols of embossed or impressed type.

3.4.1.3 Canister holders. Two canister holders shall be mounted by means of grommets on each exterior side of the facepiece in accordance with the design, location and orientation shown on 805-1327143. The grommets shall form a gastight seal between the facepiece and the canister holders. The strength of the bond shall be determined as specified in 4.6.6 and shall not be less than 50 pounds. The leakage between each canister holder and the facepiece as determined by 4.6.7 shall not exceed that permitted in the canister holder leakage requirements of MIL-M-17921.

3.4.1.4 Air deflector tubes. Two air deflector tubes in accordance with the design and location shown on 805-1327143, 805-1327140, and 805-1327141 shall be provided on the inner side of the facepiece to direct the inspired air over the lens.

3.4.1.5 Inlet valves. An inlet valve shall be installed in each canister holder. The design and assembly of the inlet valve shall be as shown on 805-1327143.

3.4.1.6 Lens. A lens in accordance with 805-1327142 shall be mounted and bonded (see Appendix A) in the facepiece to form a gastight seal. The surface of the lens in the protective mask shall be a smooth unbroken curve, free of distorted areas and blemishes as determined by the requirements for determination of physical distortion in the lens of MIL-T-19102. A distorted area is defined as a dip in excess of 1/8-inch as determined in 4.6.5.3. A 1/8-inch band around the periphery of the lens adjacent to the lens pocket shall not be included in this measurement. Scratches, creases, marks, or other defects over 1/4-inch long which would affect the wearers vision while the mask is in use shall not be permitted. The leakage of the entire seal as determined by 4.6.5.4 shall not exceed that permitted in the lens seal leakage requirements of MIL-M-17921. The bond shall not deleteriously affect the rubber or the plastic and shall not project outwards from the seal more than 1/16 inch. Any unbonded area shall not extend more than 1/16 inch inwards from the periphery of the lens opening determined by using a gage conforming with 4.6.5.5.1. The lens bonded into the facepiece, shall not break, crack, or crease when flexed as specified in 4.6.5.2. The lens, before installation in faceblank, shall have a light transmission of not less than 90 percent when tested at a wave length of 450 millimicrons (see 4.6.5.1). After installation in faceblank, and after aging, as specified in 4.6.5.3, the light transmission at this wave length shall not be less than 80 percent. Furthermore, the lens installed as above shall not show physical distortion when tested as specified in 4.6.5.3. The average strength of the lens bond both before and after aging shall not be less than 30 pounds per linear inch width and no individual specimen shall be less than 25 pounds when tested as specified in 4.6.5.5.

3.4.1.7 Speech diaphragm- outlet valve assembly. The assembly shall consist of a speech diaphragm, an outlet valve and the appurtenant parts made in accordance with 805-1327144 and 805-1327145. The assembly shall be securely fastened in the faceblank by tape and wire (or a metal strap) in a manner which will provide a gastight seal.

3.4.1.7.1 Speech diaphragm. The assembled speech diaphragm shall not fail mechanically when subjected to a pressure differential equal to 4 p.s.i. of air when tested in accordance with 4.6.9.

3.4.1.7.2 Outlet valve. The resistance of the assembled outlet valve to air flow shall not exceed 25 mm. of water when tested in accordance with 4.6.8. The leakage of this assembly when tested in a clear dry condition in accordance with 4.6.8.1 shall not exceed that permitted in the outlet valve leakage requirements of MIL-M-17921.

3.4.1.7.3 Leakage. After installation in the faceblank the leakage of the assembly as determined by 4.6.9.1 shall not exceed that permitted in speech diaphragm requirements of MIL-M-17921.

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3.4.1.8 Leakages. The sum of the leakages permitted by 3.4.1.3, 3.4.1.6 and 3.4.1.7.3, determined by 4.6.7 shall not exceed that permitted in the leakage requirements of MIL-M-17921.

3.4.1.9 Inhalation resistance. The assembled facepiece less canisters shall not have an inhalation resistance exceeding 10 mm. H_2O when tested as specified in 4.6.10.2.

3.4.1.10 Buckletabs and buckles. On the periphery of the facepiece there shall be five extensions forming tabs by which buckles are attached. The positions of these tabs and buckles on the facepiece will be known as the head strap buckle for the uppermost one, temple strap buckle for the next lower one on each side, and cheek strap buckle for each of the lowest ones. These tabs shall be formed by gradually narrowing the extension of the facepiece to the width of the buckle. The outer end of each tab shall be reinforced by laminating the rubber with fabric formed around the buckle shank. The design of the tabs, buckles and means of attachment shall be in accordance with 805-1327140,805-1327141, and 805-1327142.

3.4.1.11 Pneumatic tube leakage. The pneumatic tube of the assembled mask shall not leak when tested in accordance with 4.6.10.3.

3.4.2 Head harness. The design and dimensions of the head harness shall be in accordance with 805-1327142. The top strap and cap shall be made of a high modulus rubber meeting the requirements of table I. The temple and cheek straps shall be made of a low modulus rubber meeting the requirements of table I. The high modulus rubber from the cap shall not extend more than 3/4 inch into the low modulus straps and the low modulus rubber from the cheek and temple straps shall not extend more than 3/4 inch into the cap. On the cheek straps, these measurements shall be made from a line normal to the centerline of the strap and 7-1/4 inches from the free end. On the temple straps, these measurements shall be made from a line normal to the centerline of the strap and 6-1/2 inches from the free end. The high modulus rubber portion of the head harness shall exhibit no visible surface cracks or breaks when tested as specified in 4.6.11.

3.4.3 Carrier. The carrier shall be made in accordance with 805-1327147 and 805-1327148.

3.4.3.1 Hardware. Repairs of minor surface defects are permissible.

3.4.4 Repair parts and tools.

3.4.4.1 Repair parts. Repair parts shall be furnished as specified (see 6.2).

3.4.4.2 Wrench. The wrench (see 6.2) for assembly of the speech diaphragm shall conform to 805-1327145.

3.4.5 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. Preproduction inspection shall consist of the examination and tests specified in 4.4 and 4.5. Production shall not be started until the preproduction inspection report has been approved by the Command or agency concerned.

4.3 Quality conformance inspection.

4.3.1 Inspection lot. The supplier shall furnish written certification with each inspection lot, that he has determined (either by actual test, or through adequate control of the manufacturing process, or by certification by manufacturer), that each type of material meets the specification requirements. Certification shall contain actual test, examination, or other verifiable quality data. The proportion of each chemical element shall be shown for comparision with specification. This certification will be acceptable evidence of conformance, unless the quality conformance test in 4.6 is required and proves nonconformance.

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4.3.1.1 Rubber components. All pieces of one type molded from one batch of rubber shall be considered a lot.

4.3 1.2 Lenses. All lenses made from one process batch of the lens material shall be considered a lot.

4.3.1.3 Assembled facepieces. All facepieces assembled during one work week shall be considered a lot.

4.3.1.4 Packaged protective masks. All protective masks packaged during one work week shall be considered a lot.

4.3.1.5 Packed masks. All packed masks offered for delivery at one time shall be considered a lot.

4.3.2 Sampling for examination. A random sample of components, subassemblies, and assembled apparatus shall be selected in accordance with MIL-STD-105 at inspection level II, for the examination specified in 4.4. The acceptable quality level (AQL) shall be 2.5 for major defects and 4.0 for minor defects.

4.3.3 Sampling for quality conformance tests. A random sample of components, subassemblies, and assembled apparatus shall be selected at random in accordance with MIL-STD-105 at inspection level S1, for tests specified in 4.5. The acceptable quality level (AQL) shall be 0.25 defective.

4.4 Examination. Each of the components, subassemblies, and assembled apparatus selected in accordance with 4.3.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table II. Any item in the sample containing one or more defects shall not be offered for delivery. If the number of defective items in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

Table II - Classification of defects.

Categories:	Defects
Critical:	
1	Protective mask incomplete, component parts missing or improperly assembled.
2	Pneumatic tube punctured.
3	Outlet valve disc or seat defective or damaged.
4	Lens surface not free of cracks, splits, or holes.
Major:	
101	Rubber materials not free from pits, bumps, nonfills, porous areas, blisters, spongy areas, air traps, flow cracks, cuts, holes and tears.
102	Inlet check valve damaged or incorrectly assembled.
103	Rubber faceblank thickness not as specified.
104	Air deflector tubes orientation and wall thickness not as specified.
105	Pneumatic tube smoothness not as specified.
106	Pneumatic tube not filled under the specified atmospheric conditions, tube collapses after filling.
107	Head harness dimensions, strap locations and reinforcement not as specified.
108	Buckle tabs not reinforced as specified.
109	Facepiece buckles not operable or not as specified.
110	Speech diaphragm outlet valve assembly not as specified.
111	Canister holder not properly orientated.
112	Finish of metal surfaces not as required.
113	Mask carrier not as specified.
Minor:	
201	Metal hardware (composition) not as specified.
202	Color of rubber not as required.
203	Color of carrier not as required.

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Table II - Classification of defects (Cont'd.).

Categories	Defects
Minor	
204	Blooming of rubber not as specified.
205	Evidence of head harness reinforcement cloth exposed on flat surfaces.
206	Outlet valve cover damaged or improperly assembled.
207	Mask carrier hardware not operable.
208	Lens not bonded to face piece as specified.
209	Protective mask not clean, not free of dirt, dust, oil or grease, or extraneous foreign materials.
210	Repair parts (if required) not as specified.
211	Packaging and packing (prior to shipping) not as required.
212	Marking, characteristic data missing, incorrect, not permanent or not legible.

4.5 Quality conformance tests. Each of the sample component, subassembly and assembled apparatus selected in accordance with 4.3.3 shall be tested in accordance with 4.6 to verify compliance with this specification. Any item in the sample containing one or more defects shall not be offered for delivery. If the number of defective items in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.6 Test procedures.

4.6.1 Selection of samples. Selection of separate samples is not required for each destruction test. Samples selected in accordance with 4.3.3 may be used for a number of tests if the following order is observed:

Facepieces:

- Lens light transmission unaged.
- Lens cold flexing.
- Lens heat deformation.
- Lens light transmission after accelerated aging.
- Lens bond.
- Canister holder grommet staking.
- Pneumatic tube thickness.
- Speech diaphragm pressure test.

4.6.1.1 Samples for rubber tests. Each batch of rubber shall be evaluated for compliance with specified physical properties requirements. Ten A.S.T.M. test slabs 6 by 6 by 0.080 plus or minus 0.005 inches shall be prepared using a cure equivalent to the production cure used to manufacture the rubber component. Four additional test slabs 4 inches long by 1-1/2 inches wide and 1/2 inch thick shall be prepared. Two shall be used for the compression set test if the batch of rubber is used to mold faceblanks, canister aprons or deflector tubes and two shall be prepared for the seal aging test if the batch of rubber is to be used to mold the diaphragm gasket. Test slab preparation shall be conducted under the observation of the Government inspector.

4.6.1.2 Samples for rubber toxicity tests. Prior to the initiation of production the contractor shall provide 24 A.S.T.M. test slabs for toxicity tests required for rubber compounds used in the manufacture of the faceblank, head harness, and air deflector tubes. These samples shall be accompanied by a report giving the cure time and temperature and the quantities of the ingredients (with manufacturer's tolerances) used in the test slabs submitted for toxicity tests. Samples shall be forwarded to the Office of the Surgeon General, Department of the Army, through the contracting officer. The samples shall be marked with the following information:

(a) Samples for toxicity tests.
(b) Name of the material.
(c) Manufacturer's designation.
(d) Manufacturer's name.

Samples shall again be taken for toxicity tests whenever a major change is made in the formulation or processing.

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4.6.2 Rubber test methods. The methods specified in FED-STD-601 shall be used in conducting the tests specified in 4.6.1.1.

4.6.2.1 Oxygen bomb. The oxygen aging shall be accomplished by exposing specimens of the rubber to oxygen at 158° ± 2°F. (70° ± 1°C.), and 300 plus or minus 10 p.s.i. for 48 hours.

4.6.3 Thickness of pneumatic tube. The thickness of the contact surface of the pneumatic tube shall be measured at a minimum of ten places selected at random in each quadrant for conformance with 3.4.1.1.

4.6.4 Rubber toxicity. Such tests, including human skin tests, as are deemed necessary by the Office of the Surgeon General of the Department of the Army for properly evaluating the toxicity of any formulation will be carried out in accordance with the directions of the Surgeon General.

4.6.5 Lens.

4.6.5.1 Lens before installation in faceblank, light transmission. Light transmission of lens material shall be determined in accordance with D-1979 at a wave length of 450 millimicrons. The unaged and the aged material when interposed in the light beam shall have the values specified in 3.4.1.6.

4.6.5.2 Lens in facepiece, cold flexing. The facepiece (see 3.4.1.6) shall be placed in a cold box at a temperature of minus 29°C. ± 3°C. (-20°F. ± 5°F.) for 2 hours. Immediately upon removal, it shall be flexed by grasping each side of the pneumatic tube, midway between the temple and check tabs, and bringing the opposite sides together quickly 6 times around a mandrel 2-1/2 inches in diameter. The mandrel shall be oriented along the vertical center line of the lens and shall contact the rubber for a minimum of 1/4 inch of the lens pocket at the top and bottom of the lens.

4.6.5.3 Lens in facepiece, heat deformation. A finished facepiece, less head harness, inlet valves and canisters shall be aged in an air oven for 24 hours at 75° ± 1°C. (167°F.). Facepieces may be aged in the containers specified in the measurement of coloration and physical distortion requirements of MIL-T-19102 or on a jig capable of holding one or more masks, which meets with the approval of the Naval Ship Engineering Center. A hole, not greater than 1/4 inch diameter, may be cut in the pneumatic tube before the facepiece is put into the oven. The condition of the lens after this aging period shall be as specified in 3.4.1.6.

4.6.5.4 Lens - facepiece leakage. Facepiece leakage shall be determined as set forth in 4.6.10 for conformance to 3.4.1.6.

4.6.5.5 Lens bond strength (aged and unaged). The lens bond strength shall be determined with test specimens 1-inch wide, cut from sample facepieces. The specimens shall be approximately 1/2 rubber and 1/2 plastic and shall be cut from unaged sample facepieces and from sample facepieces aged as specified in 4.6.5.3. The strength of the bond shall be determined in a Scott testing machine, or equal, (see measurement of the lens bond strength requirements of MIL-T-19102) calibrated at 12 inches per minute travel for conformance with 3.4.1.6.

4.6.5.5.1 Gage. A gage 1/4-inch wide by 0.009 plus or minus .002-inch thick shall be used to determine depth of unbonded areas between lens and facepiece.

4.6.6 Canister holder - facepiece bond. The staking of the grommets which secure the canister holders shall be tested with the fixtures shown on D-1982 fitted on a Scott testing machine, or equal, set at a rate of travel of 12 inches per minute for conformance with 3.4.1.3. For determination, see measurement of the ND MK V canister holder to facepiece bond requirements of MIL-T-19102.

4.6.7 Canister holder grommet leakage. The canister holder grommet leakage shall be determined as specified in 4.6.10 for conformance with 3.4.1.3.

4.6.8 Outlet valve resistance. The resistance of the outlet valve assembly prior to installation in the facepiece shall be determined at a continuous flow rate of 85 l/min. for conformance with 3.4.1.7.2 using inspection apparatus E3 outlet valve resistance indicator.

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4.6.8.1 Outlet valve leakage. The outlet valve assembly shall be tested prior to installation in the mask at a pressure differential equal to 25 mm. of water for conformance with 3.4.1.7.2 using inspection apparatus M4 outlet valve leakage indicator.

4.6.9 Speech diaphragm. The assembled speech diaphragm shall be tested using inspection apparatus Q25 pressure testing device, to determine conformance with 3.4.1.7.1. For this test the protective screen shall be cut out of the completed speech diaphragm assembly. (Care shall be taken to avoid removing the periphery of the screen from its position under the retainer ring and to avoid damaging the discs.) Air pressure equal to 4 p.s.i. will then be applied to the front of the speech diaphragm at a rate which will allow a build up of approximately 1 p.s.i./sec.

4.6.9.1 Speech diaphragm-facepiece assembly leakage. The facepiece assembly leakage shall be determined as specified in 4.6.10 for conformance with 3.4.1.7.3.

4.6.10 Assembled mask, less canisters.

4.6.10.1 Leakage. The leakage of the assembled mask (without canisters) and the facepiece component parts shall be determined using inspection apparatus Q14 gas mask leakage tester. Air flow shall be adjusted to 16 liters per minute and the air supply throttled to provide a negative pressure equal to one inch of water inside the mask. These conditons shall be maintained while DOP smoke is sprayed around the lens periphery (see 3.4.1.6); canister holders (see 3.4.1.3); speech diaphragm outlet valve assembly periphery and through the guard over the speech diaphragm (see 3.4.1.7.3). During this test the outlet valve may be wet to insure no leakage. Upon completion of test the outlet valve shall be lifted to check supply of smoke and operation of smoke meter. The sum of the leakages at all points shall not exceed that specified in 3.4.1.8. For determination of leakage (see leaktesting the ND MK V facepiece requirements of MIL-T-19102).

4.6.10.2 Resistance inhalation. The inhalation resistance of the assembled facepiece, less canisters, shall be determined at a continuous air flow rate or 85 liters per minute for conformance with 3.4.1.9 using inspection apparatus Q43 inhalation resistance indicator.

4.6.10.3 Pneumatic tube. The leakage of the assembled mask (without canisters) shall be determined using the facepiece pneumatic tube leakage tester. The pneumatic tube shall not collapse or leak when subjected to a vacuum of 26 plus or minus 1 inch of mercury for 3 minutes. The mask shall be examined for leaks or collapsed tube within 3 minutes after removing the vacuum.

4.6.11 Head harness ozone resistance.

Test method	ASTM D1149
Sample	Top strap of head harness
Test temperature	90°F.
Test period	Duration of exposure of sample to specify ozone concentration: 48 hours.
Special apparatus	A jig so designed that the rubber strap may be secured at one end by a buckle, PC2-11, of 805-1327142, and at the other end by a suitable clamp. Apparatus should be designed to maintain the strap at 20 percent elongation, based upon the length of the strap prior to test.
Procedure	Thread one end of the top strap through buckle. Secure other end with clamp. Adjust apparatus to produce and maintain 20 percent elongation of strap. Subject sample to test.
Results	At the conclusion of test, sample shall exhibit no visible surface cracks, or breaks resulting from the test procedure.

4.7 Possible test failures. Possible test failures are defined as follows:

(a) Rubber compound physical properties:
  (1) Tensile stress (before and after aging) not within the required allowance when tested as specified.
  (2) Tensile strength (before and after aging) less than the minimum specified.
  (3) Elongation (before and after aging) not within the minimum value.
  (4) Hardness (before aging) not within the required tolerance.

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(5) Compression set (before aging) exceeds the specified maximum.
(6) Permanent set (before aging) exceeds the maximum allowed.
(7) Seal aging, (sealing properties) initial sealing air pressure (PSI) and aged sealing air pressure (PSI) less than the minimum specified.

(b) Rubber toxicity tests:
(1) Toxicity in excess of acceptable limits (see Government laboratory report on preproduction sample).

(c) Lens in facepiece unaged, light transmission:
(1) Light transmission less than the minimum requred when tested as specified.

(d) Lens in facepiece, cold flexing:
(1) Lens evidences cracks, breaks, or creases after cooling and flexing as specified.

(e) Lens in facepiece, heat deformation:
(1) Lens not free of distortion after testing as specified.

(f) Lens in facepiece, aged, light distoration:
(1) Light transmission less than the minimum required when tested as specified.

(g) Canister holder grommet leakage:
(1) Leakage exceeds the specified allowance.

(h) Canister holder faceblank bond:
(1) Strength of bond less than required when tested as specified.

(i) Outlet valve resistance:
(1) Resistance to air flow exceeds the allowable.

(j) Outlet valve leakage:
(1) Assembly leakage exceeds the amount allowable when tested as specified.

(k) Inhalation resistance:
(1) Inhalation resistance exceeds maximum allowed.

(l) Lens-to-faceblank bond strength:
(1) Average strength of individual specimen is less than allowable.

(m) Lens-faceblank leakage:
(1) Lens-faceblank leakage exceeds the maximum when tested as specified.

(n) Speech diaphragm pressure:
(1) Speech diaphragm fails mechanically when subjected to specified pressure.

(o) Speech diaphragm, facepiece assembly leakage:
(1) Leakage of assembly is greater than amount permitted.

(p) Assembled mask (w/o canister) leakage:
(1) Leakage exceeds maximum specified.

(q) Head harness ozone resistance.
(1) Evidence of tearing or breaking when subjected to twisting and pulling specified.

5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only to direct Government procurement. Preparation for delivery requirements between contractors and subcontractors shall be as specified in the individual order.)

5.1 Domestic shipment and early equipment use.

5.1.1 Packaging. Packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion deterioration, physical and mechanical damage during shipment from the supply source to the using activity and early use.

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal Stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.4 Onboard repair parts. Repair parts for onboard storage and future use shall be in accordance with method IC-2 of MIL-P-116. Repair parts shall be packaged Level A, packed Level C and marked Level A and C, respectively, in accordance with the applicable specification. For submarines, packaging shall be in accordance with MIL-STD-758.

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5.1.4.1 General. Rubber parts without metal fittings shall be packaged in a manner to prevent creasing, distortion, stretching or compression during storage and shipment. Rubber parts with metal fittings shall be packaged in accordance with method 1A of MIL-P-116. Items generally shall be packaged individually except when used in sets or quantities greater than one. Unit and intermediate containers shall conform to PPP-B-676, PPP-B-566, PPP-B-665, and PPP-B-636. Closure and sealing of boxes shall be in accordance with the applicable box specification.

5.1.4.2 Disc valve outlet. Outlet valve disc shall be packaged in accordance with MIL-D-10132.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.2.1.1.1 Level A.

5.2.1.1.1.1 Unit containers. Unit containers of protective masks, shall be in accordance with method IC-2 of MIL-P-116. The unit package shall contain one protective mask in a fiberboard box in accordance with PPP-B-636, type I, weather resistant class, grade 5, style RSC.

5.2.1.1.1.2 Packaging of protective mask. A tube of anti-fog shall be put into the loop provided for such in the carrier (see 6.2). A head harness shall be attached to a facepiece and the assembly held in an upright position, shall be placed in a carrier with the lens adjacent to the cotton flannel liner and directly back of the metal shield. A square of tissue paper, approximately 12 inches by 12 inches, shall be inserted between head harness and lens inner surface to prevent contact between these two components. Fold the top over the sides of the carrier folding inward and pull down to engage the snap button and fasten. The straps shall be neatly folded to lie on the back of the carrier. The entire assembly shall be placed in R.S.C. fiberboard box conforming to grade W5c of PPP-B-636, with the straps placed on the bottom and the top of the carrier placed adjacent to one end of the box. One metal package of canisters in accordance with MIL-C-24314 shall be placed in the opposite end of the box. The approximate dimension of the box should be 7 inches wide, 5-1/2 inches high and 15 inches long. Sufficient cushioning materials conforming to PPP-C-843 shall be placed inside the corrugated box to prevent movement of the carrier and the metal can during shipment and handling. The amount of cushioning material used shall not be sufficient to distort the carrier.

5.2.1.1.2 Level C.

5.2.1.1.2.1 Unit containers. Unit containers of protective masks, shall be in accordance with method III of MIL-P-116. The unit package shall contain one protective mask.

5.2.2 Packing. Packing shall be level A or B as specified (see 6.2).

5.2.2.1 Level A.

5.2.2.1.1 Exterior containers. Protective masks, packaged as specified (see 6.2), shall be packed in containers conforming to PPP-B-640 and PPP-B-636. All center and edge seams and manufacturer's joints are to be sealed and waterproofed with pressure sensitive tape in accordance with applicable box specification. Shipping containers shall be closed and banded in accordance with the applicable box specification or appendix thereto. The gross weight shall not exceed the weight limitations of the applicable box specification.

5.2.2.2 Level B.

5.2.2.2.1 Exterior containers. Protective masks, packaged as specified (see 6.2), shall be packed in containers conforming to PPP-B-640 and PPP-B-636. Shipping containers shall be closed in accordance with the applicable box specification or appendix thereto. The gross weight shall not exceed the weight limitations of the applicable box specification.

5.2.3 Index list. An index list of repair parts conforming to MIL-P-15137 shall be inserted in the repair parts shipping containers.)

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5.2.4 Marking.

5.2.4.1 General. In addition to any special marking requirements, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2.4.2 Special. The canister lot number(s) shall be shown on the exterior of the canister package and the protective mask package (see 5.2.1.1.1.2) and on the shipping container (see 5.2.2).

6. NOTES

6.1 Intended use. The protective masks covered by this specification are intended for use for respiratory protection of personnel aboard ships against radiological, biological and chemical warfare. It is not designed to give protection against gasoline vapors, refrigerants, carbon monoxide fumes and other industrial gases.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color, if different (see 3.3.1.4).
(c) Number of spare parts required.
(d) Number of wrenches required (see 3.4.4.2).
(e) Repair parts required (see 3.4.4.1).
(f) Whether anti-fog kit is required and how supplied.
(g) That contractors furnished formulation for the following and certify:
    (1) The faceblanks furnished are nonsulphurblooming and comply with the specified rubber content.
    (2) The names and quantities of antioxidants and accelerators used for compounding the rubber
    (3) The names and quantities of sun checking retardant, copper inhibitor, and coloring materials used.
(h) The size of the manufacturer's lot.
(i) Packaging, packing, and marking requirements other than those required by 5.1 (see 5.2).

6.3 Color chips. Color chips may be obtained upon application to the Laboratory Officer, Test Laboratory Building 121, Philadelphia Naval Shipyard, Naval Base Station, Philadelphia, Pa. 19112. When requesting, specify for what purpose chips are desired.

6.4 Preproduction.

6.4.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 4240-N344)

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10. SCOPE

10.1 This appendix describes the procedures for bonding lens and formulation of cement used in bonding lens into gas mask.

10.2 Surface preparation. The finished lens shall be degreased with a solvent such as acetone, or preferably, a chlorinated solvent such as perchloroethylene (Perclene) and allowed to air dry. The rubber shall be washed thoroughly with a 1 percent solution of detergent (cold water ALL) to remove stearates and any other water dispersible contaminants and allowed to air dry. The channel should then be wiped with "Perclene" to remove any residual silicones or grease and allowed to air dry.

10.3 Cement components. The cement components used for bonding lens shall be as follows:

(a) Component A (100 Pbw)
Dupont "Adiprene" L-100 (reaction product of a diisocyanate and a polyalkylene ether glycol).
(b) Component B (55 Pbw)
Xylene (1.4 dimethyl benzene) $C_6H_4(CH_3)_2$.
(c) Component C (12.5 Pbw)
Dupont "Moca" (4,4' Methylene (2 chloroaniline).

10.4 Formulation of cement. Mix component A and B at room temperature. Component C (which melts at about 200°F. (100°C.) is heated to 250°F. (do not heat above 250°F.). Good ventilation must be provided for mixing and heating component C. Component C at 250°F. should then be mixed with the mixture of components A and B with agitation. Care should be taken to minimize mixing in air bubbles. The resulting mixture has a pot life of about 30 minutes at 70°F., and shorter at higher temperatures.

10.4 Procedure. Place mask on a mounting jig and insert clean dry lens into mask, taking care not to touch the bond area. A bead of adhesive is applied around the channel so that all mating surfaces have a thin film. Avoid local over-application which might run onto the visible part of the lens when the jig is clamped. The jig is closed which applies only contact pressure uniformly around the facepiece. The mask is allowed to set followed by a cure of one hour at 200°F.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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